

# Girraween High School

## Mathematics

Year 11

HSC Task 1, 2010

Time Allowed : 90 minutes

Examiner: Mr. Chris Howard

Instructions : Attempt all questions.

Start each question on a new sheet of paper .

Use only one side of paper (with no columns).

All necessary working **must** be shown.

Marks may be deducted for careless or badly arranged work.

16

Question 1. (15 Marks)

- (a) Find the locus of the point  $P(x,y)$  that moves so that
- (i) It is equidistant from  $A(-1, -2)$  and  $B(3, 1)$  3
  - (ii) It is equidistant from the point  $S(2, 3)$  and the line  $y = -2$  3
  - (iii) Its distance from  $A(-1, -2)$  is twice its distance from  $B(5, 1)$  3
  - (iv) Describe fully and sketch your locus in part (iii) 3 4
- (b) Find the equation of the parabola with focus  $(0, a)$  and directrix  $y = -a$  3 4 3  
(show all working)

Question 2 (16 Marks)

- (a) Express the equation of the circle  $x^2 - 4x + y^2 + 8y + 11 = 0$  in the form  $(x - l)^2 + (y - k)^2 = r^2$ . Hence state the radius and centre. 4
- (b) Sketch the graph of  $y^2 = -4x$  clearly indicating the focus and the directrix. 4
- (c) Find the equation of the curve which is always 3 units from  $(2, -3)$  2
- (d) Find the equation/s of the curve which is always 3 units from  $x = 2$  2
- (e) Find the vertex, focal length, focus and the directrix of the parabola  $(x - 2)^2 = 8(y - 1)$  6

Question 3. (13 Marks)

For the parabola  $y = \frac{1}{4}(x^2 + 6x + 5)$  find

- (i) The equation of the tangent at (3 , 8) 3
- (ii) The equation of the normal at (1 , 3) 3
- (iii) The coordinates of the vertex 2
- (iv) Find the coordinates of the focus 2
- (v) Find the equation of the focal chord passing through (1 , 3) 3

Question 4 (16 Marks)

(a) For the arithmetic progression 2, 5, 8, 11, 14 .... Find

- (i) a and d 2
- (ii) The formula for the  $n^{\text{th}}$  term 2
- (iii) The  $20^{\text{th}}$  term 2
- (iv) The sum of the first 20 terms 3

(b) The sum of an Arithmetic Progression is given by  $S_n = n^2 + 2n$

- (i) Find the first four terms of the A.P. 3
- (ii) Find the sum of the first 20 terms 2
- (iii) Find the  $20^{\text{th}}$  term 2

Question 5 (16 Marks)

(a) For the geometric progression 1000, 900, 810, 729, ..... Find

- (i) The common ratio 1
- (ii) The  $10^{\text{th}}$  term 2
- (iii) The sum of the first 10 terms 2
- (iv) The limiting sum of the series. 2

(b) The  $3^{\text{rd}}$  term of a G.P. is 12 the  $5^{\text{th}}$  term is 24

- (i) Find the two common ratios. 3
- (ii) Find the first term 2
- (iii) Find the value of the sum of the first 8 terms for both values of r  
leave your answer as a rationalised surd. 4

Question 6 (23 Marks)

(a) Evaluate

(i)  $\sum_{n=1}^{10} 3n+1$  3      (ii)  $\sum_{n=3}^{10} 3(2^n)$  3

(b) The numbers  $a, b, 18$  are in Arithmetic Progression the numbers  $24, b, a$  are in Geometric progression find all possible values for  $a$  and  $b$  5

(c)  $P(x, y)$  is a point on the locus which is equidistant from the point  $S(3, 1)$  and the line  $y = x$ .

(i) Find an expression for the distance  $PS$ . 2

(ii) Using the perpendicular distance formula find an expression for the perpendicular distance from  $y = x$  to  $P$  3

Hint  $P_d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$

(iii) Find the equation of the locus. 3

(iv) Find the points of intersection with the  $x$  axis 2

(v) Prove the locus never intersects the  $y$ -axis. 2

HSC ASS. TASK 1  
DEC. 2010.

Q1(a) PA = PB

$$\sqrt{(x+1)^2 + (y+2)^2} = \sqrt{(x-3)^2 + (y-1)^2}$$

$$(x+1)^2 + (y+2)^2 = (x-3)^2 + (y-1)^2$$

$$x^2 + 2x + 1 + y^2 + 4y + 4 = x^2 - 6x + 9 + y^2 - 2y + 1$$

$$8x + 5 = -6y + 10$$

$$8x + 6y - 5 = 0$$

$$\text{R } y = -\frac{4}{3}x + \frac{5}{6} \quad (3)$$

ii) PS = PM

$$\sqrt{(x-2)^2 + (y-3)^2} = y + 2$$

$$(x-2)^2 + (y-3)^2 = (y+2)^2$$

$$x^2 - 4x + 4 + y^2 - 6y + 9 = y^2 + 4y + 4$$

$$(x-2)^2 = 10y - 5$$

$$(x-2)^2 = 5(2y-1) \quad (3)$$

iii) PA = 2PB

$$\sqrt{(x+1)^2 + (y+2)^2} = 2\sqrt{(x-5)^2 + (y-1)^2}$$

$$(x+1)^2 + (y+2)^2 = 4\{(x-5)^2 + (y-1)^2\}$$

$$x^2 + 2x + 1 + y^2 + 4y + 4$$

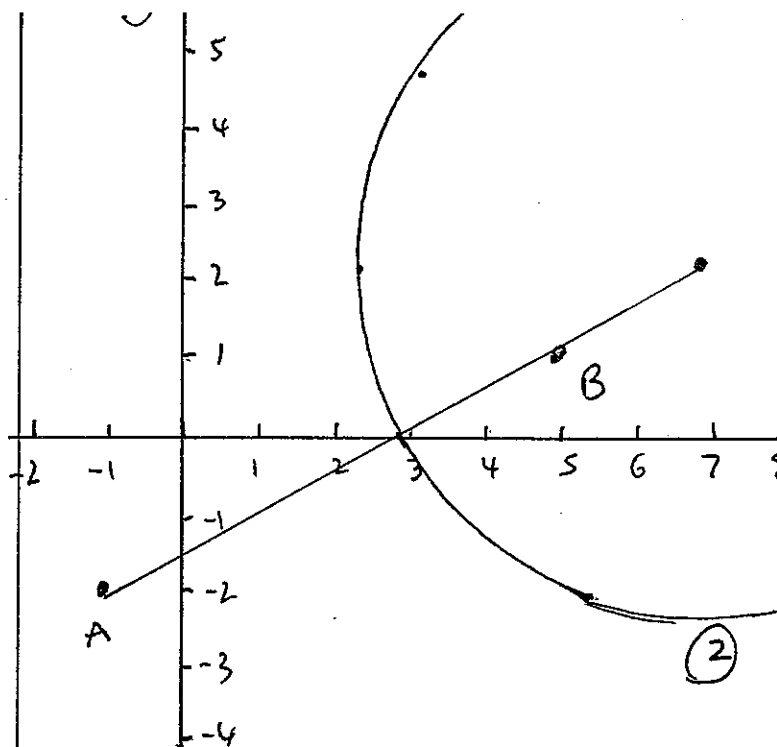
$$= 4x^2 - 40x + 100 + 4y^2 - 8y + 4$$

$$0 = 3x^2 - 42x + 99 + 3y^2 - 12y$$

$$0 = x^2 - 14x + y^2 - 4y + 33$$

$$20 = (x^2 - 14x + 49) + (y^2 - 4y + 4)$$

$$20 = (x-7)^2 + (y-2)^2$$



CIRCLE CENTRE (7, 2).

RADIUS ( $\sqrt{20}$ ). (2)

(b) PS = PM

$$\sqrt{x^2 + (y-a)^2} = y + a$$

$$x^2 + y^2 - 2ay + a^2 = y^2 + 2ay + a^2$$

$$x^2 = 4ay. \quad (3)$$

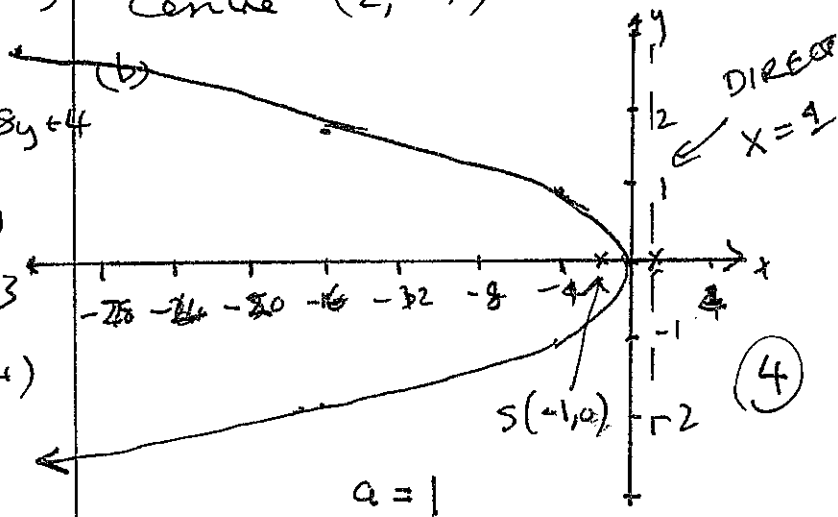
Question 2.

$$(a) x^2 - 4x + y^2 + 8y = -11$$

$$x^2 - 4x + 4 + y^2 + 8y + 16 = -11 + 20$$

$$(x-2)^2 + (y+4)^2 = 9. \quad (4)$$

Centre (2, -4) Radius 3.



$$(c) \quad (x-2)^2 + (y+3)^2 = 9 \quad (2)$$

$$(d) \quad x = 5 \text{ or } x = -1 \quad (2)$$

$$(e) \quad (x-2)^2 = 8(y-1)$$

$$\text{VERTEX } (2, 1) \quad (1)$$

$$\text{FOCAL LENGTH IS } 2 \quad (1)$$

$$\text{FOCUS } (2, 3) \quad (2)$$

$$\text{DIRECTRIX } y = -1 \quad (2)$$

### Question 3

$$(i) \quad y = \frac{1}{4}(x^2 + 6x + 5)$$

$$\frac{dy}{dx} = \frac{1}{4}(2x + 6) = \frac{1}{2}(x + 3)$$

$$\frac{dy}{dx}(x=1) = \frac{1}{2}(3+3) = 3$$

EQ<sup>n</sup> OF TANGENT

$$y - y_1 = m(x - x_1)$$

$$y - 8 = 3(x - 3)$$

$$y - 8 = 3x - 9$$

$$\boxed{y = 3x - 1} \quad (3)$$

(ii) GRADIENT OF TANGENT AT (1, 3)

$$\frac{dy}{dx}(x=1) = \frac{1}{2}(1+3) = 2$$

$$\therefore \text{GRADIENT OF NORMAL} = -\frac{1}{2} \quad (m_1 m_2 = -1)$$

$$y - 3 = -\frac{1}{2}(x - 1)$$

$$y - 3 = -\frac{x}{2} + \frac{1}{2}$$

$$y = -\frac{x}{2} + \frac{7}{2} \quad (3)$$

(iii) Coordinates of Vertex

$$\frac{dy}{dx} = 0$$

$$0 = \frac{1}{2}(x+3)$$

$$0 = x + 3$$

$$\underline{x = -3}$$

$$\text{Point } V(-3, -1) \quad (2)$$

$$(iv) \quad 4y + 4 = x^2 + 6x + 9$$

$$4(y+1) = (x+3)^2$$

$$\text{FOCAL LENGTH } a = 1$$

$$\therefore \text{FOCUS } (-3, 0) \quad (2)$$

$$(v) \quad S(-3, 0) \quad P(1, 3)$$

$$\therefore m_{SP} = \frac{3-0}{1+3} = \frac{3}{4}$$

$\therefore$  EQ<sup>n</sup> SP

$$y - 3 = \frac{3}{4}(x - 1)$$

$$y - 3 = \frac{3}{4}x - \frac{3}{4}$$

$$\boxed{y = \frac{3}{4}x + \frac{9}{4}} \quad (3)$$

$$(i) \quad a = 2 \quad d = 3 \quad (2)$$

$$(ii) \quad T_n = a + (n-1)d$$

$$T_n = 2 + (n-1)3 \quad (2)$$

$$(iii) \quad T_{20} = 2 + (19)3$$

$$= 59 \quad (2)$$

$$(iv) \quad S_{20} = \frac{n}{2}(a + l)$$

$$= \frac{20}{2}(2 + 59)$$

$$(11) \quad (2)$$

### Question 4

$$(b) (i) S_n = n^2 + 2n$$

$$S_1 = 3 = T_1$$

$$S_2 = 8 = T_1 + T_2$$

$$\therefore T_2 = 5$$

$$S_3 = 15 \therefore T_3 = 7$$

$$S_4 = 24 \therefore T_4 = 9$$

$$3, 5, 7, 9 \dots$$

(3)

$$(ii) S_{20} = 20^2 + 2(20)$$

$$= 440$$

(2)

$$(iii) T_{20} = S_{20} - S_{19}$$

$$= 440 - 399$$

$$= 41$$

(2)

### Question 5.

$$(a) (i) r = 0.9$$

(1)

$$(ii) T_{10} = a(r)^9$$

$$= 387.42$$

(2)

$$(iii) S_{10} = \frac{a(r^{10} - 1)}{r - 1}$$

$$S_{10} = \frac{1000(1 - 0.9^{10})}{0.1}$$

$$= 6513.22$$

(2)

$$(iv) S_{\infty} = \frac{a}{1-r}$$

$$= \frac{1000}{0.1}$$

$$= 10,000$$

(2)

$$(1) (i) T_3 = ar^4$$

$$ar^2 = 12$$

$$T_5 = ar^4$$

$$ar^4 = 24$$

$$\frac{24}{12} = \frac{ar^4}{ar^2}$$

(3)

$$r^2 = 2 \quad r = \pm\sqrt{2}$$

$$(ii) ar^2 = 12$$

$$2a = 12$$

$$a = 6$$

(2)

$$(iii) S_n = \frac{a(r^n - 1)}{r - 1}$$

$$S_8 = \frac{6(\sqrt{2}^8 - 1)}{\sqrt{2} - 1}$$

$$= \frac{6(15)}{\sqrt{2} - 1} \times \frac{\sqrt{2} + 1}{\sqrt{2} + 1}$$

$$= 90(\sqrt{2} + 1)$$

(2)

$$\text{OR } S_8 = \frac{6((- \sqrt{2})^8 - 1)}{-\sqrt{2} - 1}$$

$$= \frac{-90}{\sqrt{2} + 1} \times \frac{\sqrt{2} - 1}{\sqrt{2} - 1}$$

$$= 90(1 - \sqrt{2})$$

(2)

# Question 6

$$(a) (i) \sum_{n=1}^{10} 3n+1 = 4+7+10+13 \dots 31$$

A.P.

$$S_n = \frac{n}{2}(a+l)$$

$$= \frac{10}{2}(4+31)$$

$$= 175. \quad (3)$$

$$(ii) \sum_{n=3}^{10} 3(2^n) = 24+72+216 + \dots + 3072$$

$$S_8 = \frac{a(r^n-1)}{r-1}$$

$$= \frac{24(2^8-1)}{1}$$

$$= 6120 \quad (3)$$

(b) A.P.

$$\therefore 18-b = b-a$$

$$18+a = 2b \quad (A)$$

G.P.

$$\frac{b}{24} = \frac{a}{b}$$

$$b^2 = 24a \quad (B)$$

$$\therefore b^2 = 24(2b-18)$$

$$b^2 - 48b + 432 = 0$$

$$(b-36)(b-12) = 0$$

$$\therefore b = 12, 36$$

$$\therefore a = 6, 54 \quad (5)$$

$$(c) (i) PS = \sqrt{(x-3)^2 + (y-1)^2} \quad (2)$$

$$(ii) y=x \Rightarrow x-y+0=0$$

$$\therefore a=1 \quad b=-1 \quad c=0$$

$$P_d = \frac{|1x-1y+0|}{\sqrt{1^2+(-1)^2}}$$

$$= \frac{|x-y|}{\sqrt{2}} \quad (3)$$

$$(iii) PS = P_d$$

$$\sqrt{(x-3)^2 + (y-1)^2} = \frac{|x-y|}{\sqrt{2}}$$

$$2\{(x-3)^2 + (y-1)^2\} = (x-y)^2$$

$$2x^2 - 12x + 18 + 2y^2 - 4y + 2 = x^2 - 2xy + y^2$$

$$x^2 - 12x + y^2 - 4y + 2xy + 20 = 0 \quad (3)$$

$$(iv) \text{ x axis } y=0$$

$$x^2 - 12x + 20 = 0$$

$$(x-10)(x-2) = 0$$

$$x = 2, 10. \quad (2)$$

$$(v) \text{ y axis } x=0$$

$$y^2 - 4y + 20 = 0$$

$$y = \frac{4 \pm \sqrt{16-80}}{2} \leftarrow \text{no solution}$$

$$\Delta < 0. \quad (2)$$